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What is Parametric and Non-Parametric in Statistic? Learn with Python Example

The common word in the data field that people often miss

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While learning Data Science, I am sure you will encounter many unfamiliar terms. Most of the statistical word that exists in the book or online courses would mention the word with little proper explanation.

One of the terms I often hear without elaborate definition is **Parametric** and **Non-Parametric**. This word was used in many Data Science learning modules, but most of the time just brushed away. Furthermore, statistics often have the same terms that refer to different things.

I want to explain Parametric and Non-Parametric terms in this article, especially in data science and statistics. Let's get into it.

Parametric Statistics

When you search the Parametric Statistics on the Internet, the first thing shown is the Parametric Statistic Method. So, what is the parametric method?

We can describe the parametric method as a **statistical method class that uses specific probability distribution with fixed parameters**. It means the method relies on the distribution assumptions made of the data.

For example, a t-test is a test to assess if the means between the group is significant or not. As the test would compare both the group means, the test would assume that the data follows a normal distribution. Deviation from this assumption might undermine the test result.

Let's use a sample dataset to use the t-test. I would use the titanic dataset from the Seaborn package for this sample.

```
#Import necessary packages
import pandas as pd
```

```
import seaborn as sns
from scipy.stats import ttest_ind, shapiro

titanic = sns.load_dataset('titanic')

#We would divide the group into two according to the survival
#and we want to compare the fare

not_survived_fare = titanic[titanic['survived'] == 0]['fare']
survived_fare = titanic[titanic['survived'] == 1]['fare']
```

Before we try to do the t-test, we want to see if the data follow the normal distribution. There are a few ways to assess it, but we can use the Shapiro-Wilk test as one of the techniques. Let's see if the not surviving fare follows the normal distribution.

```
# Test the normality of the not survived fare data
stat, p = stats.shapiro(not_survived_fare)
print('Statistics=%.3f, p=%.3f' % (stat, p))
alpha = 0.05
if p < alpha:
    print('Fail to Reject H0 (Data distributed Normal)')
else:
    print('Reject H0 (Data not distributed Normal)')
```

```
Statistics=0.513, p=0.000
Fail to Reject H0 (Data distributed Normal)
```

Image by Author

The data follows the normal distribution; how about the other dataset group?

```
stat, p = stats.shapiro(survived_fare)
print('Statistics=%.3f, p=%.3f' % (stat, p))
alpha = 0.05
if p < alpha:
    print('Fail to Reject H0 (Data distributed Normal)')
else:
    print('Reject H0 (Data not distributed Normal)')
```

```
Statistics=0.597, p=0.000  
Fail to Reject H0 (Data distributed Normal)
```

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It seems the data also follow the normal distribution. Let's do a t-test to see if there is a significant difference between both group's means.

```
t_stat, p = ttest_ind(survived_fare, not_survived_fare)  
  
print('Statistics=%.3f, p=%.3f' % (t_stat, p))  
alpha = 0.05  
if p < alpha:  
    print('Fail to Reject H0 (Significant)')  
else:  
    print('Reject H0 (Not-Significant)')
```

```
Statistics=7.939, p=0.000  
Fail to Reject H0 (Significant)
```

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There are significant differences between the fare, as the t-test shows the evidence.

There are various examples of Parametric Statistic, including:

1. ANOVA (Analysis of Variance)
2. Linear regression
3. Logistic regression
4. Pearson correlation

Every single one of them relies on specific data distribution. That is why they are considered Parametric.

Non-Parametric Statistics

In contrast with Parametric statistics, the non-parametric statistic does not make assumptions regarding the underlying data distribution. It does not estimate population parameters and uses ranking or other non-parametric measures. It doesn't matter if the data follow a normal distribution or not; the reliability is still the same.

Non-parametric is more robust to outliers because the technique makes no assumption.

But, parametric tests were more powerful because the data followed the assumed distribution, and to achieve the same power level as a parametric test, a non-parametric might require a larger sample size. Additionally, the non-parametric was not as effective as the parametric test in detecting slight differences between the groups.

Nevertheless, the usefulness of the non-parametric test is still great when the data distribution violates the parametric test assumption.

Let's take an example of a non-parametric test Mann-Whitney U. It's the non-parametric test equivalent of a t-test. Mann-Whitney U test is based on the data ranking rather than the data parameter itself.

Let's use the Python example to understand better.

```
from scipy.stats import mannwhitneyu

# perform Mann-Whitney U test
u_stat, p = mannwhitneyu(not_survived_fare, survived_fare)

print('Statistics=%.3f, p=%.3f' % (u_stat, p))
alpha = 0.05
if p < alpha:
    print('Fail to Reject H0 (Significant)')
else:
    print('Reject H0 (Not-Significant)')
```

```
Statistics=57806.500, p=0.000  
Fail to Reject H0 (Significant)
```

Image by Author

There are still significant differences between the two groups, as shown in the result above.

Non-parametric test techniques include the following but are not limited to:

1. Wilcoxon signed-rank test
2. Kruskal-Wallis test
3. Friedman test
4. Spearman's rank correlation coefficient
5. Kendall's tau correlation coefficient

Summary

In summary, parametric statistics assume a specific distribution and estimate the population parameters. In contrast, non-parametric statistics have no assumption to the data distribution and use non-parametric measurement (such as ranking).

Parametric statistics could be more powerful if we met the assumptions, but violations were easier to disturb the assumption. On the other hand, non-parametric statistics are more robust to violations but less powerful when the parametric assumptions are met.

I hope it helps!

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